analysis_final.out

```
Mplus VERSION 8.11
MUTHEN & MUTHEN
04/16/2026 7:19 PM

INPUT INSTRUCTIONS


  TITLE: Bayesian Analysis - Reference Group White Female;

  DATA:
    FILE IS mplus_input_data.dat;

  VARIABLE:
    NAMES ARE REGN RSESN SPED;

    ! Category 13 (White_F) is removed from here to make it the reference
    USEVARIABLES ARE SPED
      AFAM_F AFAM_M AMIND_F AMIND_M ASIAN_F ASIAN_M
      HAW_F HAW_M HISP_F HISP_M MULTI_F MULTI_M WHT_M;

    CATEGORICAL IS SPED;
    MISSING ARE ALL (-99);

  DEFINE:
    AFAM_F = 0; IF (REGN EQ 1) THEN AFAM_F = 1;
    AFAM_M = 0; IF (REGN EQ 2) THEN AFAM_M = 1;
    AMIND_F = 0; IF (REGN EQ 3) THEN AMIND_F = 1;
    AMIND_M = 0; IF (REGN EQ 4) THEN AMIND_M = 1;
    ASIAN_F = 0; IF (REGN EQ 5) THEN ASIAN_F = 1;
    ASIAN_M = 0; IF (REGN EQ 6) THEN ASIAN_M = 1;
    HAW_F = 0; IF (REGN EQ 7) THEN HAW_F = 1;
    HAW_M = 0; IF (REGN EQ 8) THEN HAW_M = 1;
    HISP_F = 0; IF (REGN EQ 9) THEN HISP_F = 1;
    HISP_M = 0; IF (REGN EQ 10) THEN HISP_M = 1;
    MULTI_F = 0; IF (REGN EQ 11) THEN MULTI_F = 1;
    MULTI_M = 0; IF (REGN EQ 12) THEN MULTI_M = 1;
    WHT_M = 0; IF (REGN EQ 14) THEN WHT_M = 1;
    ! REGN 13 (White_Female) is now the implicit reference group.

  ANALYSIS:
    ESTIMATOR = BAYES;
    PROCESSORS = 4;
    BITERATIONS = (5000);
    BCONVERGENCE = 0.01;

  MODEL:
  ! Model now strictly tests group differences against white females
    SPED ON AFAM_F AFAM_M AMIND_F AMIND_M ASIAN_F ASIAN_M
            HAW_F HAW_M HISP_F HISP_M MULTI_F MULTI_M WHT_M;

  OUTPUT:
    TECH1 TECH8 STDYX;




INPUT READING TERMINATED NORMALLY



Bayesian Analysis - Reference Group White Female;

SUMMARY OF ANALYSIS
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```
Number of groups 1
Number of observations 1389102

Number of dependent variables 1
Number of independent variables 13
Number of continuous latent variables 0

Observed dependent variables

  Binary and ordered categorical (ordinal)
   SPED

Observed independent variables
   AFAM_F AFAM_M AMIND_F AMIND_M ASIAN_F ASIAN_M
   HAW_F HAW_M HISP_F HISP_M MULTI_F MULTI_M
   WHT_M


Estimator BAYES
Specifications for Bayesian Estimation
  Point estimate MEDIAN
  Number of Markov chain Monte Carlo (MCMC) chains 2
  Random seed for the first chain 0
  Starting value information UNPERTURBED
  Algorithm used for Markov chain Monte Carlo GIBBS(PX1)
  Convergence criterion 0.100D-01
  Maximum number of iterations 50000
  K-th iteration used for thinning 1
Link PROBIT

Input data file(s)
  mplus_input_data.dat
Input data format FREE


SUMMARY OF DATA



COVARIANCE COVERAGE OF DATA

Minimum covariance coverage value 0.100

     Number of missing data patterns 1


     PROPORTION OF DATA PRESENT


           Covariance Coverage
              SPED AFAM_F AFAM_M AMIND_F AMIND_M
              ________ ________ ________ ________ ________
 SPED 1.000
 AFAM_F 1.000 1.000
 AFAM_M 1.000 1.000 1.000
 AMIND_F 1.000 1.000 1.000 1.000
 AMIND_M 1.000 1.000 1.000 1.000 1.000
 ASIAN_F 1.000 1.000 1.000 1.000 1.000
 ASIAN_M 1.000 1.000 1.000 1.000 1.000
 HAW_F 1.000 1.000 1.000 1.000 1.000
 HAW_M 1.000 1.000 1.000 1.000 1.000
 HISP_F 1.000 1.000 1.000 1.000 1.000
 HISP_M 1.000 1.000 1.000 1.000 1.000
 MULTI_F 1.000 1.000 1.000 1.000 1.000
 MULTI_M 1.000 1.000 1.000 1.000 1.000
 WHT_M 1.000 1.000 1.000 1.000 1.000
```

Covariance Coverage

	ASIAN_F	ASIAN_M	HAW_F	HAW_M	HISP_F
ASIAN_F	1.000				
ASIAN_M	1.000	1.000			
HAW_F	1.000	1.000	1.000		
HAW_M	1.000	1.000	1.000	1.000	
HISP_F	1.000	1.000	1.000	1.000	1.000
HISP_M	1.000	1.000	1.000	1.000	1.000
MULTI_F	1.000	1.000	1.000	1.000	1.000
MULTI_M	1.000	1.000	1.000	1.000	1.000
WHT_M	1.000	1.000	1.000	1.000	1.000

Covariance Coverage

	HISP_M	MULTI_F	MULTI_M	WHT_M
HISP_M	1.000			
MULTI_F	1.000	1.000		
MULTI_M	1.000	1.000	1.000	
WHT_M	1.000	1.000	1.000	1.000

UNIVARIATE PROPORTIONS AND COUNTS FOR CATEGORICAL VARIABLES

SPED

Category 1	0.838	1164590.000
Category 2	0.162	224512.000

UNIVARIATE SAMPLE STATISTICS

UNIVARIATE HIGHER-ORDER MOMENT DESCRIPTIVE STATISTICS

Variable/ Sample Size	Mean/ Variance	Skewness/ Kurtosis	Minimum/ Maximum	% with Min/Max	Percentiles 20%/60%	40%/80%	Median
AFAM_F	0.149	1.967	0.000	85.06%	0.000	0.000	0.000
1389102.000	0.127	1.870	1.000	14.94%	0.000	0.000	
AFAM_M	0.155	1.904	0.000	84.48%	0.000	0.000	0.000
1389102.000	0.131	1.626	1.000	15.52%	0.000	0.000	
AMIND_F	0.002	22.516	0.000	99.80%	0.000	0.000	0.000
1389102.000	0.002	504.951	1.000	0.20%	0.000	0.000	
AMIND_M	0.002	21.772	0.000	99.79%	0.000	0.000	0.000
1389102.000	0.002	472.029	1.000	0.21%	0.000	0.000	
ASIAN_F	0.020	6.942	0.000	98.05%	0.000	0.000	0.000
1389102.000	0.019	46.197	1.000	1.95%	0.000	0.000	
ASIAN_M	0.021	6.738	0.000	97.93%	0.000	0.000	0.000
1389102.000	0.020	43.403	1.000	2.07%	0.000	0.000	
HAW_F	0.001	39.075	0.000	99.93%	0.000	0.000	0.000
1389102.000	0.001	1524.849	1.000	0.07%	0.000	0.000	
HAW_M	0.001	38.423	0.000	99.93%	0.000	0.000	0.000
1389102.000	0.001	1474.343	1.000	0.07%	0.000	0.000	
HISP_F	0.087	2.933	0.000	91.31%	0.000	0.000	0.000
1389102.000	0.079	6.602	1.000	8.69%	0.000	0.000	
HISP_M	0.090	2.870	0.000	91.02%	0.000	0.000	0.000
1389102.000	0.082	6.235	1.000	8.98%	0.000	0.000	
MULTI_F	0.021	6.694	0.000	97.91%	0.000	0.000	0.000
1389102.000	0.020	42.816	1.000	2.09%	0.000	0.000	
MULTI_M	0.021	6.685	0.000	97.90%	0.000	0.000	0.000
1389102.000	0.021	42.690	1.000	2.10%	0.000	0.000	
WHT_M	0.224	1.326	0.000	77.63%	0.000	0.000	0.000
1389102.000	0.174	-0.241	1.000	22.37%	0.000	1.000	

```
THE MODEL ESTIMATION TERMINATED NORMALLY

     USE THE FBITERATIONS OPTION TO INCREASE THE NUMBER OF ITERATIONS BY A FACTOR
     OF AT LEAST TWO TO CHECK CONVERGENCE AND THAT THE PSR VALUE DOES NOT INCREASE.



MODEL FIT INFORMATION

Number of Free Parameters 14

Bayesian Posterior Predictive Checking using Chi-Square

          95% Confidence Interval for the Difference Between
          the Observed and the Replicated Chi-Square Values

                              -14.265 15.527

          Posterior Predictive P-Value 0.520



MODEL RESULTS

                                Posterior One-Tailed 95% C.I.
                    Estimate S.D. P-Value Lower 2.5% Upper 2.5% Significance

 SPED ON
    AFAM_F 0.179 0.005 0.000 0.170 0.187 *
    AFAM_M 0.569 0.004 0.000 0.561 0.578 *
    AMIND_F 0.041 0.031 0.106 -0.023 0.099
    AMIND_M 0.385 0.027 0.000 0.333 0.437 *
    ASIAN_F -0.467 0.013 0.000 -0.493 -0.441 *
    ASIAN_M -0.057 0.011 0.000 -0.078 -0.037 *
    HAW_F -0.101 0.059 0.044 -0.220 0.013
    HAW_M 0.158 0.051 0.001 0.056 0.259 *
    HISP_F 0.033 0.006 0.000 0.021 0.044 *
    HISP_M 0.365 0.005 0.000 0.356 0.376 *
    MULTI_F 0.024 0.010 0.009 0.004 0.044 *
    MULTI_M 0.376 0.009 0.000 0.358 0.393 *
    WHT_M 0.373 0.004 0.000 0.364 0.381 *

 Thresholds
    SPED$1 1.246 0.003 0.000 1.240 1.252 *


STANDARDIZED MODEL RESULTS


STDYX Standardization

                                Posterior One-Tailed 95% C.I.
                    Estimate S.D. P-Value Lower 2.5% Upper 2.5% Significance
 SPED ON
    AFAM_F 0.062 0.002 0.000 0.059 0.065 *
    AFAM_M 0.201 0.001 0.000 0.198 0.204 *
    AMIND_F 0.002 0.001 0.106 -0.001 0.004
    AMIND_M 0.017 0.001 0.000 0.015 0.019 *
    ASIAN_F -0.063 0.002 0.000 -0.067 -0.060 *
    ASIAN_M -0.008 0.001 0.000 -0.011 -0.005 *
    HAW_F -0.003 0.001 0.044 -0.005 0.000
    HAW_M 0.004 0.001 0.001 0.001 0.007 *
    HISP_F 0.009 0.002 0.000 0.006 0.012 *
    HISP_M 0.102 0.001 0.000 0.099 0.105 *
    MULTI_F 0.003 0.001 0.009 0.001 0.006 *
    MULTI_M 0.053 0.001 0.000 0.050 0.055 *
```

WHT_M 0.151 0.002 0.000 0.148 0.155 *

Thresholds
SPED$1 1.215 0.003 0.000 1.209 1.221 *

R-SQUARE

	Posterior		One-Tailed	95% C.I.	
Variable	Estimate	S.D.	P-Value	Lower 2.5%	Upper 2.5%
SPED	0.049	0.001	0.000	0.048	0.050

TECHNICAL 1 OUTPUT

PARAMETER SPECIFICATION

TAU

SPED$1
14

NU

SPED	AFAM_F	AFAM_M	AMIND_F	AMIND_M
0	0	0	0	0

NU

ASIAN_F	ASIAN_M	HAW_F	HAW_M	HISP_F
0	0	0	0	0

NU

HISP_M	MULTI_F	MULTI_M	WHT_M
0	0	0	0

LAMBDA

	SPED	AFAM_F	AFAM_M	AMIND_F	AMIND_M
SPED	0	0	0	0	0
AFAM_F	0	0	0	0	0
AFAM_M	0	0	0	0	0
AMIND_F	0	0	0	0	0
AMIND_M	0	0	0	0	0
ASIAN_F	0	0	0	0	0
ASIAN_M	0	0	0	0	0
HAW_F	0	0	0	0	0
HAW_M	0	0	0	0	0
HISP_F	0	0	0	0	0
HISP_M	0	0	0	0	0
MULTI_F	0	0	0	0	0
MULTI_M	0	0	0	0	0
WHT_M	0	0	0	0	0

LAMBDA

	ASIAN_F	ASIAN_M	HAW_F	HAW_M	HISP_F
SPED	0	0	0	0	0

```
AFAM_F 0 0 0 0 0
AFAM_M 0 0 0 0 0
AMIND_F 0 0 0 0 0
AMIND_M 0 0 0 0 0
ASIAN_F 0 0 0 0 0
ASIAN_M 0 0 0 0 0
HAW_F 0 0 0 0 0
HAW_M 0 0 0 0 0
HISP_F 0 0 0 0 0
HISP_M 0 0 0 0 0
MULTI_F 0 0 0 0 0
MULTI_M 0 0 0 0 0
WHT_M 0 0 0 0 0


          LAMBDA
             HISP_M MULTI_F MULTI_M WHT_M
             ________ ________ ________ ________
SPED 0 0 0 0
AFAM_F 0 0 0 0
AFAM_M 0 0 0 0
AMIND_F 0 0 0 0
AMIND_M 0 0 0 0
ASIAN_F 0 0 0 0
ASIAN_M 0 0 0 0
HAW_F 0 0 0 0
HAW_M 0 0 0 0
HISP_F 0 0 0 0
HISP_M 0 0 0 0
MULTI_F 0 0 0 0
MULTI_M 0 0 0 0
WHT_M 0 0 0 0


          THETA
             SPED AFAM_F AFAM_M AMIND_F AMIND_M
             ________ ________ ________ ________ ________
SPED 0
AFAM_F 0 0
AFAM_M 0 0 0
AMIND_F 0 0 0 0
AMIND_M 0 0 0 0 0
ASIAN_F 0 0 0 0 0
ASIAN_M 0 0 0 0 0
HAW_F 0 0 0 0 0
HAW_M 0 0 0 0 0
HISP_F 0 0 0 0 0
HISP_M 0 0 0 0 0
MULTI_F 0 0 0 0 0
MULTI_M 0 0 0 0 0
WHT_M 0 0 0 0 0


          THETA
             ASIAN_F ASIAN_M HAW_F HAW_M HISP_F
             ________ ________ ________ ________ ________
ASIAN_F 0
ASIAN_M 0 0
HAW_F 0 0 0
HAW_M 0 0 0 0
HISP_F 0 0 0 0 0
HISP_M 0 0 0 0 0
MULTI_F 0 0 0 0 0
MULTI_M 0 0 0 0 0
WHT_M 0 0 0 0 0
```

```
          THETA
             HISP_M MULTI_F MULTI_M WHT_M
             ________ ________ ________ ________
HISP_M 0
MULTI_F 0 0
MULTI_M 0 0 0
WHT_M 0 0 0 0


          ALPHA
             SPED AFAM_F AFAM_M AMIND_F AMIND_M
             ________ ________ ________ ________ ________
                   0 0 0 0 0


          ALPHA
             ASIAN_F ASIAN_M HAW_F HAW_M HISP_F
             ________ ________ ________ ________ ________
                   0 0 0 0 0


          ALPHA
             HISP_M MULTI_F MULTI_M WHT_M
             ________ ________ ________ ________
                   0 0 0 0


          BETA
             SPED AFAM_F AFAM_M AMIND_F AMIND_M
             ________ ________ ________ ________ ________
SPED 0 1 2 3 4
AFAM_F 0 0 0 0 0
AFAM_M 0 0 0 0 0
AMIND_F 0 0 0 0 0
AMIND_M 0 0 0 0 0
ASIAN_F 0 0 0 0 0
ASIAN_M 0 0 0 0 0
HAW_F 0 0 0 0 0
HAW_M 0 0 0 0 0
HISP_F 0 0 0 0 0
HISP_M 0 0 0 0 0
MULTI_F 0 0 0 0 0
MULTI_M 0 0 0 0 0
WHT_M 0 0 0 0 0


          BETA
             ASIAN_F ASIAN_M HAW_F HAW_M HISP_F
             ________ ________ ________ ________ ________
SPED 5 6 7 8 9
AFAM_F 0 0 0 0 0
AFAM_M 0 0 0 0 0
AMIND_F 0 0 0 0 0
AMIND_M 0 0 0 0 0
ASIAN_F 0 0 0 0 0
ASIAN_M 0 0 0 0 0
HAW_F 0 0 0 0 0
HAW_M 0 0 0 0 0
HISP_F 0 0 0 0 0
HISP_M 0 0 0 0 0
MULTI_F 0 0 0 0 0
MULTI_M 0 0 0 0 0
WHT_M 0 0 0 0 0


          BETA
             HISP_M MULTI_F MULTI_M WHT_M
```

```
                    ________ ________ ________ ________
SPED 10 11 12 13
AFAM_F 0 0 0 0
AFAM_M 0 0 0 0
AMIND_F 0 0 0 0
AMIND_M 0 0 0 0
ASIAN_F 0 0 0 0
ASIAN_M 0 0 0 0
HAW_F 0 0 0 0
HAW_M 0 0 0 0
HISP_F 0 0 0 0
HISP_M 0 0 0 0
MULTI_F 0 0 0 0
MULTI_M 0 0 0 0
WHT_M 0 0 0 0


          PSI
             SPED AFAM_F AFAM_M AMIND_F AMIND_M
             ________ ________ ________ ________ ________
SPED 0
AFAM_F 0 0
AFAM_M 0 0 0
AMIND_F 0 0 0 0
AMIND_M 0 0 0 0 0
ASIAN_F 0 0 0 0 0
ASIAN_M 0 0 0 0 0
HAW_F 0 0 0 0 0
HAW_M 0 0 0 0 0
HISP_F 0 0 0 0 0
HISP_M 0 0 0 0 0
MULTI_F 0 0 0 0 0
MULTI_M 0 0 0 0 0
WHT_M 0 0 0 0 0


          PSI
             ASIAN_F ASIAN_M HAW_F HAW_M HISP_F
             ________ ________ ________ ________ ________
ASIAN_F 0
ASIAN_M 0 0
HAW_F 0 0 0
HAW_M 0 0 0 0
HISP_F 0 0 0 0 0
HISP_M 0 0 0 0 0
MULTI_F 0 0 0 0 0
MULTI_M 0 0 0 0 0
WHT_M 0 0 0 0 0


          PSI
             HISP_M MULTI_F MULTI_M WHT_M
             ________ ________ ________ ________
HISP_M 0
MULTI_F 0 0
MULTI_M 0 0 0
WHT_M 0 0 0 0


    STARTING VALUES


          TAU
             SPED$1
             ________
               0.915
```

NU

SPED	AFAM_F	AFAM_M	AMIND_F	AMIND_M
0.000	0.000	0.000	0.000	0.000

NU

ASIAN_F	ASIAN_M	HAW_F	HAW_M	HISP_F
0.000	0.000	0.000	0.000	0.000

NU

HISP_M	MULTI_F	MULTI_M	WHT_M
0.000	0.000	0.000	0.000

LAMBDA

	SPED	AFAM_F	AFAM_M	AMIND_F	AMIND_M
SPED	1.000	0.000	0.000	0.000	0.000
AFAM_F	0.000	1.000	0.000	0.000	0.000
AFAM_M	0.000	0.000	1.000	0.000	0.000
AMIND_F	0.000	0.000	0.000	1.000	0.000
AMIND_M	0.000	0.000	0.000	0.000	1.000
ASIAN_F	0.000	0.000	0.000	0.000	0.000
ASIAN_M	0.000	0.000	0.000	0.000	0.000
HAW_F	0.000	0.000	0.000	0.000	0.000
HAW_M	0.000	0.000	0.000	0.000	0.000
HISP_F	0.000	0.000	0.000	0.000	0.000
HISP_M	0.000	0.000	0.000	0.000	0.000
MULTI_F	0.000	0.000	0.000	0.000	0.000
MULTI_M	0.000	0.000	0.000	0.000	0.000
WHT_M	0.000	0.000	0.000	0.000	0.000

LAMBDA

	ASIAN_F	ASIAN_M	HAW_F	HAW_M	HISP_F
SPED	0.000	0.000	0.000	0.000	0.000
AFAM_F	0.000	0.000	0.000	0.000	0.000
AFAM_M	0.000	0.000	0.000	0.000	0.000
AMIND_F	0.000	0.000	0.000	0.000	0.000
AMIND_M	0.000	0.000	0.000	0.000	0.000
ASIAN_F	1.000	0.000	0.000	0.000	0.000
ASIAN_M	0.000	1.000	0.000	0.000	0.000
HAW_F	0.000	0.000	1.000	0.000	0.000
HAW_M	0.000	0.000	0.000	1.000	0.000
HISP_F	0.000	0.000	0.000	0.000	1.000
HISP_M	0.000	0.000	0.000	0.000	0.000
MULTI_F	0.000	0.000	0.000	0.000	0.000
MULTI_M	0.000	0.000	0.000	0.000	0.000
WHT_M	0.000	0.000	0.000	0.000	0.000

LAMBDA

	HISP_M	MULTI_F	MULTI_M	WHT_M
SPED	0.000	0.000	0.000	0.000
AFAM_F	0.000	0.000	0.000	0.000
AFAM_M	0.000	0.000	0.000	0.000
AMIND_F	0.000	0.000	0.000	0.000
AMIND_M	0.000	0.000	0.000	0.000
ASIAN_F	0.000	0.000	0.000	0.000
ASIAN_M	0.000	0.000	0.000	0.000

HAW_F	0.000	0.000	0.000	0.000
HAW_M	0.000	0.000	0.000	0.000
HISP_F	0.000	0.000	0.000	0.000
HISP_M	1.000	0.000	0.000	0.000
MULTI_F	0.000	1.000	0.000	0.000
MULTI_M	0.000	0.000	1.000	0.000
WHT_M	0.000	0.000	0.000	1.000

THETA

	SPED	AFAM_F	AFAM_M	AMIND_F	AMIND_M
SPED	0.000				
AFAM_F	0.000	0.000			
AFAM_M	0.000	0.000	0.000		
AMIND_F	0.000	0.000	0.000	0.000	
AMIND_M	0.000	0.000	0.000	0.000	0.000
ASIAN_F	0.000	0.000	0.000	0.000	0.000
ASIAN_M	0.000	0.000	0.000	0.000	0.000
HAW_F	0.000	0.000	0.000	0.000	0.000
HAW_M	0.000	0.000	0.000	0.000	0.000
HISP_F	0.000	0.000	0.000	0.000	0.000
HISP_M	0.000	0.000	0.000	0.000	0.000
MULTI_F	0.000	0.000	0.000	0.000	0.000
MULTI_M	0.000	0.000	0.000	0.000	0.000
WHT_M	0.000	0.000	0.000	0.000	0.000

THETA

	ASIAN_F	ASIAN_M	HAW_F	HAW_M	HISP_F
ASIAN_F	0.000				
ASIAN_M	0.000	0.000			
HAW_F	0.000	0.000	0.000		
HAW_M	0.000	0.000	0.000	0.000	
HISP_F	0.000	0.000	0.000	0.000	0.000
HISP_M	0.000	0.000	0.000	0.000	0.000
MULTI_F	0.000	0.000	0.000	0.000	0.000
MULTI_M	0.000	0.000	0.000	0.000	0.000
WHT_M	0.000	0.000	0.000	0.000	0.000

THETA

	HISP_M	MULTI_F	MULTI_M	WHT_M
HISP_M	0.000			
MULTI_F	0.000	0.000		
MULTI_M	0.000	0.000	0.000	
WHT_M	0.000	0.000	0.000	0.000

ALPHA

SPED	AFAM_F	AFAM_M	AMIND_F	AMIND_M
0.000	0.000	0.000	0.000	0.000

ALPHA

ASIAN_F	ASIAN_M	HAW_F	HAW_M	HISP_F
0.000	0.000	0.000	0.000	0.000

ALPHA

HISP_M	MULTI_F	MULTI_M	WHT_M
0.000	0.000	0.000	0.000

BETA

	SPED	AFAM_F	AFAM_M	AMIND_F	AMIND_M
SPED	0.000	0.000	0.000	0.000	0.000
AFAM_F	0.000	0.000	0.000	0.000	0.000
AFAM_M	0.000	0.000	0.000	0.000	0.000
AMIND_F	0.000	0.000	0.000	0.000	0.000
AMIND_M	0.000	0.000	0.000	0.000	0.000
ASIAN_F	0.000	0.000	0.000	0.000	0.000
ASIAN_M	0.000	0.000	0.000	0.000	0.000
HAW_F	0.000	0.000	0.000	0.000	0.000
HAW_M	0.000	0.000	0.000	0.000	0.000
HISP_F	0.000	0.000	0.000	0.000	0.000
HISP_M	0.000	0.000	0.000	0.000	0.000
MULTI_F	0.000	0.000	0.000	0.000	0.000
MULTI_M	0.000	0.000	0.000	0.000	0.000
WHT_M	0.000	0.000	0.000	0.000	0.000

BETA

	ASIAN_F	ASIAN_M	HAW_F	HAW_M	HISP_F
SPED	0.000	0.000	0.000	0.000	0.000
AFAM_F	0.000	0.000	0.000	0.000	0.000
AFAM_M	0.000	0.000	0.000	0.000	0.000
AMIND_F	0.000	0.000	0.000	0.000	0.000
AMIND_M	0.000	0.000	0.000	0.000	0.000
ASIAN_F	0.000	0.000	0.000	0.000	0.000
ASIAN_M	0.000	0.000	0.000	0.000	0.000
HAW_F	0.000	0.000	0.000	0.000	0.000
HAW_M	0.000	0.000	0.000	0.000	0.000
HISP_F	0.000	0.000	0.000	0.000	0.000
HISP_M	0.000	0.000	0.000	0.000	0.000
MULTI_F	0.000	0.000	0.000	0.000	0.000
MULTI_M	0.000	0.000	0.000	0.000	0.000
WHT_M	0.000	0.000	0.000	0.000	0.000

BETA

	HISP_M	MULTI_F	MULTI_M	WHT_M
SPED	0.000	0.000	0.000	0.000
AFAM_F	0.000	0.000	0.000	0.000
AFAM_M	0.000	0.000	0.000	0.000
AMIND_F	0.000	0.000	0.000	0.000
AMIND_M	0.000	0.000	0.000	0.000
ASIAN_F	0.000	0.000	0.000	0.000
ASIAN_M	0.000	0.000	0.000	0.000
HAW_F	0.000	0.000	0.000	0.000
HAW_M	0.000	0.000	0.000	0.000
HISP_F	0.000	0.000	0.000	0.000
HISP_M	0.000	0.000	0.000	0.000
MULTI_F	0.000	0.000	0.000	0.000
MULTI_M	0.000	0.000	0.000	0.000
WHT_M	0.000	0.000	0.000	0.000

PSI

	SPED	AFAM_F	AFAM_M	AMIND_F	AMIND_M
SPED	1.000				
AFAM_F	0.000	0.064			
AFAM_M	0.000	0.000	0.066		
AMIND_F	0.000	0.000	0.000	0.001	
AMIND_M	0.000	0.000	0.000	0.000	0.001

```
 ASIAN_F 0.000 0.000 0.000 0.000 0.000
 ASIAN_M 0.000 0.000 0.000 0.000 0.000
 HAW_F 0.000 0.000 0.000 0.000 0.000
 HAW_M 0.000 0.000 0.000 0.000 0.000
 HISP_F 0.000 0.000 0.000 0.000 0.000
 HISP_M 0.000 0.000 0.000 0.000 0.000
 MULTI_F 0.000 0.000 0.000 0.000 0.000
 MULTI_M 0.000 0.000 0.000 0.000 0.000
 WHT_M 0.000 0.000 0.000 0.000 0.000


           PSI
              ASIAN_F ASIAN_M HAW_F HAW_M HISP_F
              ________ ________ ________ ________ ________
 ASIAN_F 0.010
 ASIAN_M 0.000 0.010
 HAW_F 0.000 0.000 0.000
 HAW_M 0.000 0.000 0.000 0.000
 HISP_F 0.000 0.000 0.000 0.000 0.040
 HISP_M 0.000 0.000 0.000 0.000 0.000
 MULTI_F 0.000 0.000 0.000 0.000 0.000
 MULTI_M 0.000 0.000 0.000 0.000 0.000
 WHT_M 0.000 0.000 0.000 0.000 0.000


           PSI
              HISP_M MULTI_F MULTI_M WHT_M
              ________ ________ ________ ________
 HISP_M 0.041
 MULTI_F 0.000 0.010
 MULTI_M 0.000 0.000 0.010
 WHT_M 0.000 0.000 0.000 0.087



     PRIORS FOR ALL PARAMETERS PRIOR MEAN PRIOR VARIANCE PRIOR STD. DEV.

     Parameter 1~N(0.000,5.000) 0.0000 5.0000 2.2361
     Parameter 2~N(0.000,5.000) 0.0000 5.0000 2.2361
     Parameter 3~N(0.000,5.000) 0.0000 5.0000 2.2361
     Parameter 4~N(0.000,5.000) 0.0000 5.0000 2.2361
     Parameter 5~N(0.000,5.000) 0.0000 5.0000 2.2361
     Parameter 6~N(0.000,5.000) 0.0000 5.0000 2.2361
     Parameter 7~N(0.000,5.000) 0.0000 5.0000 2.2361
     Parameter 8~N(0.000,5.000) 0.0000 5.0000 2.2361
     Parameter 9~N(0.000,5.000) 0.0000 5.0000 2.2361
     Parameter 10~N(0.000,5.000) 0.0000 5.0000 2.2361
     Parameter 11~N(0.000,5.000) 0.0000 5.0000 2.2361
     Parameter 12~N(0.000,5.000) 0.0000 5.0000 2.2361
     Parameter 13~N(0.000,5.000) 0.0000 5.0000 2.2361
     Parameter 14~N(0.000,5.000) 0.0000 5.0000 2.2361


TECHNICAL 8 OUTPUT


   TECHNICAL 8 OUTPUT FOR BAYES ESTIMATION

     PROCESSOR BSEED
     1 0
     2 285380
     3 253358
     4 93468

                     POTENTIAL PARAMETER WITH
     ITERATION SCALE REDUCTION HIGHEST PSR
     100 1.099 4
```

```
     200 1.061 11
     300 1.022 6
     400 1.075 7
     500 1.013 6
     600 1.012 5
     700 1.014 5
     800 1.027 7
     900 1.026 6
     1000 1.031 6
     1100 1.020 6
     1200 1.021 7
     1300 1.018 7
     1400 1.006 7
     1500 1.014 3
     1600 1.015 3
     1700 1.009 3
     1800 1.007 3
     1900 1.005 5
     2000 1.002 3
     2100 1.003 3
     2200 1.003 3
     2300 1.006 3
     2400 1.009 7
     2500 1.007 7
     2600 1.003 7
     2700 1.001 6
     2800 1.003 6
     2900 1.002 6
     3000 1.001 6
     3100 1.001 5
     3200 1.001 5
     3300 1.002 6
     3400 1.003 5
     3500 1.002 5
     3600 1.004 6
     3700 1.003 5
     3800 1.006 5
     3900 1.004 5
     4000 1.003 5
     4100 1.002 5
     4200 1.003 5
     4300 1.003 5
     4400 1.005 5
     4500 1.004 5
     4600 1.003 5
     4700 1.003 5
     4800 1.002 5
     4900 1.003 6
     5000 1.003 5


DIAGRAM INFORMATION

  Use View Diagram under the Diagram menu in the Mplus Editor to view the diagram.
  If running Mplus from the Mplus Diagrammer, the diagram opens automatically.

  Diagram output
    s:\ibrahim\uconn spencer project analysis delaware data\analysis_final.dgm

     Beginning Time: 19:19:23
        Ending Time: 19:44:06
       Elapsed Time: 00:24:43
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